

Vinay Yerram

Senior AI/ML Engineer

Phone: 945-200-6860 || Mail: vinay.y1278@gmail.com

LinkedIn: <https://www.linkedin.com/in/y-vinay/>



Professional Summary:

- **Senior AI/ML Engineer with 11+ years of end-to-end engineering experience**, successfully evolving from building distributed big data pipelines to architecting production-grade machine learning and **Generative AI** systems.
- Deep domain expertise across the **Financial Services** and **Healthcare** sectors, delivering high-impact, scalable systems for **AML anomaly detection**, credit-risk scoring, **clinical NLP**, and patient readmission predictions.
- Architected and deployed complex **multi-agent AI architectures** utilizing **LangGraph** and **LangChain** to automate multi-step reasoning workflows, clinical decision-support routing, and regulatory compliance reviews.
- Designed enterprise-scale **RAG pipelines** leveraging high-performance vector databases such as **Pinecone** and **Milvus**, utilizing semantic chunking and embeddings to ground responses across clinical guidelines and banking policies.
- Fine-tuned and deployed open-weight foundation models, including **Llama 3**, utilizing **Hugging Face PEFT (QLoRA)** for highly specialized NLP classification tasks while optimizing cloud GPU inference costs.
- Implemented advanced prompt engineering strategies and rigorously evaluated hallucination rates and retrieval relevance utilizing the **RAGAS** and **Langfuse** evaluation frameworks to ensure enterprise AI safety.
- Developed high-performance predictive classification pipelines utilizing **XGBoost** and **LightGBM**, applying advanced hyperparameter tuning to maximize **ROC-AUC** and drastically reduce false positives in fraud and risk models.
- Engineered advanced time-series forecasting models and deep learning architectures leveraging **LSTM** and **Prophet** to accurately predict banking liquidity demands and hospital emergency department resource utilization.
- Built robust distributed data processing workflows and feature engineering pipelines utilizing **Apache Spark**, **PySpark**, and **Kafka**, processing terabytes of streaming financial telemetry and **EHR** (Electronic Health Record) data.
- Standardized **MLOps** and model governance practices across organizations utilizing **Amazon SageMaker**, **Vertex AI**, and **MLflow**, enforcing strict version lineage and automated **CI/CD** deployment workflows.
- Established proactive machine learning observability utilizing **Evidently AI** to continuously track data drift, prediction skew, and performance degradation in dynamic, real-world production environments.
- Containerized and deployed low-latency ML inference microservices utilizing **FastAPI**, Docker, **Amazon EKS**, and **Azure Kubernetes Service (AKS)** to support high-throughput, real-time predictive APIs.
- Integrated model explainability frameworks including **SHAP** and enforced strict **PII/PHI data masking** to maintain regulatory compliance for both HIPAA and international banking data governance standards.

Technology Stack:

Generative AI & Agents	Agentic Workflows (LangGraph, CrewAI), MCP, RAG, LLM Fine-tuning (QLoRA, DeepSpeed), Prompt Engineering
ML/Deep Learning	PyTorch, TensorFlow, Scikit-learn, XGBoost, Neural Networks (CNN, RNN), Hyperparameter Tuning (Optuna)
Data & Cloud Infra	AWS (SageMaker, Bedrock, Glue), Azure (ML, ADF), Snowflake, Redshift, Spark/PySpark, Kafka, SQL/NoSQL
MLOps & Production	End-to-End Pipelines (Kubeflow, MLflow), Model Monitoring (Evidently AI, Langfuse), CI/CD (GitHub Actions)
Vector Search & NLP	Qdrant, Pinecone, FAISS, ColBERT, Hugging Face Transformers (BERT, T5, GPT), Tokenization
Architectural Ops	Kubernetes (EKS/AKS), Docker, Triton Inference Server, vLLM, ONNX, FastAPI, Microservices

Professional Experience:

Roles & Responsibilities:

- Architected enterprise cloud-native AI infrastructure utilizing **AWS SageMaker**, **Amazon Bedrock**, and **AWS Glue**, establishing scalable MLOps pipelines for quantitative research and algorithmic risk modeling that reduced new-model time-to-market by 45%.
- Led the development of an internal wealth-management GenAI Copilot utilizing **Llama 3** via **Amazon Bedrock**, serving financial advisors with portfolio insights and driving a 25% boost in response relevance for client inquiries.
- Designed multi-agent AI systems utilizing **LangGraph** to automate complex **KYC/AML** (Know Your Customer / Anti-Money Laundering) workflows, slashing manual compliance review time by 50% and reclaiming ~3,500 billable hours annually.
- Engineered advanced prompt-engineering workflows utilizing chain-of-thought (CoT) and dynamic few-shot prompting, optimizing **Llama 3** reasoning for complex financial document summarization and sentiment analysis on quarterly earnings transcripts.
- Engineered a hybrid **RAG pipeline** utilizing **Pinecone** for vector search across 100,000+ SEC filings and internal regulatory documents, improving retrieval latency by 40% and boosting semantic search precision by 30%.
- Fine-tuned domain-specific **Llama 3** models utilizing **Hugging Face PEFT (QLoRA)** and distributed training on **AWS SageMaker**, cutting inference costs by 40% while maintaining high accuracy for proprietary financial NLP classification tasks.
- Optimized large language model serving architectures utilizing **vLLM** on production GPU clusters, achieving a 3x throughput increase and a 40% reduction in token generation latency for real-time advisory applications.
- Developed high-performance predictive engines utilizing **XGBoost** and **PyTorch**, handling 10M+ daily predictions for real-time credit-default scoring and transaction fraud detection with 35% lower inference latency.
- Implemented **LSTM** neural networks for multivariate time-series forecasting, capturing complex temporal patterns in market data to improve branch-level liquidity forecasting and optimize daily cash reserves.
- Scaled distributed data ingestion pipelines utilizing **Python**, **PySpark**, and **Apache Airflow**, reliably processing 1.2B+ daily financial transaction records for integration with **Amazon Redshift** while reducing ETL compute overhead by 18%.
- Established standardized model deployment lifecycles utilizing **MLflow** for experiment tracking and **Feast** as a unified feature store, compressing the deployment lifecycle of risk models from weeks to under 4 days.
- Containerized and orchestrated high-concurrency RESTful machine learning APIs utilizing **FastAPI** and **Amazon EKS**, supporting 1,000+ concurrent requests for live transaction scoring with a strict 99.99% uptime SLA.
- Implemented automated generative evaluation frameworks utilizing **RAGAS** and **Langfuse**, driving down hallucination rates by 40% and ensuring strict output compliance with enterprise financial safety policies.
- Built real-time ML observability platforms utilizing **Evidently AI** for automated concept-drift detection in fraud models, and enforced strict **PII masking** guardrails to protect sensitive customer financial data.

Responsibilities:

- Optimized large-scale batch data processing utilizing **PySpark** on **Amazon EMR**, engineering reliable ETL pipelines to process millions of daily **EHR** (Electronic Health Record) events while reducing cloud compute costs by 30%.
- Built production-grade feature engineering frameworks utilizing **Pandas** and **Scikit-learn**, standardizing clinical data imputation and reducing data preparation time by 25% for downstream ML workflows.
- Developed predictive classification models utilizing **Random Forest** and **Optuna** for hyperparameter tuning, successfully identifying high-risk patients for 30-day readmission with an 18% improvement in predictive accuracy.
- Implemented time-series forecasting solutions utilizing **Prophet** to predict emergency department utilization and patient inflow, improving resource allocation efficiency and model generalization by 20%.
- Designed deep learning architectures utilizing **PyTorch** to perform clinical NLP tasks, extracting critical diagnosis codes and symptoms from unstructured physician notes to streamline medical coding processes.
- Engineered an enterprise **RAG pipeline** utilizing **Pinecone** for vector search and **OpenAI text-embedding-ada-002** for semantic chunking, boosting retrieval relevance across thousands of clinical guidelines by 30%.
- Architected generative AI applications integrating **GPT-4** via the **OpenAI SDK**, assisting clinical support staff with policy retrieval and improving conversational query resolution rates by 35%.

- Developed sequential agentic workflows leveraging **LangChain** to automate the triage of incoming patient symptom reports, intelligently routing high-severity cases to triage nurses and cutting manual review time.
- Applied advanced prompt engineering techniques, including dynamic few-shot prompting and clinical safety guardrails, to ensure **GPT-4** generated medically accurate and compliant summaries of complex patient histories.
- Fine-tuned domain-specific NLP models utilizing **Hugging Face Transformers**, adapting open-source architectures to securely classify sensitive medical documents directly within the enterprise cloud network.
- Containerized machine learning inference workloads utilizing Docker and deployed scalable microservices on **Amazon EKS** using **FastAPI**, reducing inference latency by 35% while maintaining 99.9% system uptime for real-time applications.
- Engineered automated **MLOps** pipelines utilizing **MLflow** for experiment tracking, hyperparameter logging, and model registry, compressing the deployment lifecycle for critical healthcare predictive models.
- Automated CI/CD and continuous retraining pipelines using **Apache Airflow** and **AWS CodePipeline**, accelerating release cycles by 40% and mitigating model degradation incidents in production.
- Implemented strict HIPAA-compliant cloud security architectures utilizing **AWS Secrets Manager**, **Amazon VPC**, and **AWS IAM** to enforce **PHI/PII** masking and ensure enterprise-grade security for all healthcare data.

Client: Royal Bank of Canada, New York City, NY

Jul 2017 - Jul 2020

Machine Learning Engineer

Responsibilities:

- Engineered scalable data processing workflows using **Apache Spark** and **Kafka**, integrating with **Azure Data Factory** to manage the daily ingestion of 3TB+ of retail banking transactions for batch and real-time fraud detection.
- Built robust feature engineering pipelines utilizing **Python**, **Pandas**, and **NumPy**, optimizing data transformation processes to generate rolling-window transaction features and reducing data preparation time by 30%.
- Developed high-performance predictive models utilizing **XGBoost** and **Scikit-learn** within **Azure Machine Learning** to score retail credit risk, applying **GridSearchCV** to improve default prediction accuracy by 14%.
- Designed and trained **RNN** (Recurrent Neural Network) architectures utilizing **TensorFlow** and **Keras** to model sequential credit card transaction behaviors, improving the detection of synthetic identity fraud.
- Implemented **LSTM** networks for multivariate time-series forecasting, capturing seasonal spending patterns to predict branch-level cash liquidity demands with a 15% reduction in **RMSE**.
- Orchestrated complex ML and ETL data workflows utilizing **Apache Airflow**, ensuring fault tolerance and automated scheduling for nightly batch-scoring workloads across millions of customer accounts.
- Applied **K-Means** clustering to segment customer spending profiles and transaction behaviors, enabling the retail banking team to deliver highly targeted financial product recommendations.
- Containerized machine learning models utilizing Docker and deployed them as low-latency microservices using **Flask** and **Gunicorn** on **Azure Kubernetes Service (AKS)** to support real-time loan approval APIs.
- Optimized system-level performance on Linux environments through parallelization and resource tuning, reducing real-time inference latency to sub-150ms for synchronous banking applications.
- Monitored production model performance using **Azure Monitor**, implementing proactive retraining strategies to mitigate data and concept drift caused by shifting macroeconomic environments.
- Transformed raw model outputs into actionable financial insights, developing interactive BI dashboards utilizing **Tableau** to communicate portfolio risk metrics clearly to non-technical banking stakeholders.
- Architected reproducible ML lifecycles and enforced engineering standards, managing version control via Git and automating **CI/CD** deployment pipelines utilizing **Jenkins** for zero-downtime releases.
- Established isolated model training environments using Conda and integrated automated data-quality validation checks to ensure zero missing values in production risk-modeling datasets.
- Enforced strict enterprise data governance protocols, applying data masking to sensitive **PII** (Personally Identifiable Information) before model training to maintain strict compliance with Canadian banking regulations.

Client: Target, Minneapolis, MN

Jan 2016 - Jun 2017

Data/ Python Engineer

Responsibilities:

- Built modular ETL frameworks using **Python 3.6**, **Apache Spark** and **Azure Data Factory**, implementing custom **Python** logic to automate complex data movement and achieving 99.9% pipeline reliability.

- Engineered distributed processing modules in **PySpark**, leveraging **Spark's** lazy evaluation and memory management to transform multi-terabyte datasets on **Azure HDInsight**.
- Developed event-driven **Python** workflows using **Azure Functions** and **Cosmos DB**, utilizing asynchronous programming to validate and ingest high-velocity streaming data in real-time.
- Built secure, high-concurrency data access layers using **Flask** and **SQLAlchemy ORM**, optimizing database interactions through **Python**-based migration scripts and advanced query tuning in **PostgreSQL**.
- Utilized the **Pandas** and **NumPy** ecosystems to engineer high-efficiency data cleaning and normalization routines, applying vectorized operations to replace slow iterative loops.
- Partnered with Data Science teams to wrap predictive models in custom **Python** production wrappers, ensuring seamless real-time inference and model versioning using **Scikit-learn**.
- Streamlined the deployment of **Python**-based microservices by containerizing environments with **Docker** and orchestrating lifecycles via **Azure Kubernetes Service (AKS)**.
- Led code integrity by developing comprehensive test suites with **Pytest**, utilizing mocks and fixtures to maintain high coverage for mission-critical data transformations.
- Integrated **Python** build processes into **Jenkins** and **Git** pipelines, automating linting (**Flake8/Black**) and deployment to reduce manual overhead and ensure **PEP 8** compliance.

Client: **Cheniere Energy, Houston, TX**

Dec 2013 - Dec 2015

Python Developer

Responsibilities:

- Designed and maintained robust back-end applications using **Django**, ensuring seamless integration with legacy **Python** environments.
- Built dynamic, responsive user interfaces using **HTML5**, **CSS3** and **AngularJS** to manage DOM manipulation and client-side logic.
- Managed relational data structures using **MySQL**, leveraging **SQLAlchemy** and the **Django ORM** to write optimized queries and handle data migrations.
- Developed and documented scalable APIs using **Django REST Framework** to facilitate communication between the front-end and back-end services.
- Implemented background job processing and scheduled tasks using **Celery** with **RabbitMQ** as a message broker to improve application performance.
- Ensured full-stack reliability by architecting comprehensive test suites, leveraging **PyTest** for backend logic and **Selenium** with **Jasmine** for automated UI and regression testing.
- Configured and managed application environments using **Virtualenv** and **Fabric**, deploying applications to **Nginx** servers using **Gunicorn**.
- Collaborated in **Agile** environments using **Jira** for sprint tracking, while maintaining code quality through structured **Git** branching and peer reviews.

Certifications:

- AWS Certified Machine Learning Specialty
- AWS Certified AI Practitioner Foundational
- Microsoft Certified AI Engineer Associate

Education:

- Bachelors in Computer Science.